

Prasanna Maddila

PHD CANDIDATE | MARL & GAME THEORY

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Education

[University of Toulouse](#) | [Lab MIAT, INRAE](#)

Toulouse, France

PHD CANDIDATE

Oct 2023 - Present

- Working on the Nash Equilibrium computation in games of Teams vs. Independent Adversaries, with application to Anti-Poaching.
- Implemented GPU-compatible solvers in JAX for proposed algorithms, and RL environment for application case.
- Project funded by the French-German ANR Project CHIP-GT.

CentraleSupélec

Paris, France

MASTERS IN ENGINEERING (COMPUTER SCIENCE)

August 2021 - 2023

- Specialisation in Theoretical Computer Science, and part of the Research Track (*Filière de Recherche*).
- Student in the Accelerated Engineering Degree Programme.

Mahindra University

Hyderabad, India

BACHELORS' IN TECHNOLOGY, MECHANICAL ENGINEERING

August 2017 - 2021

- Recipient of the University's Student Scholarship for Academic Excellence for 3 years.
- Member of the Organising Team TEDx (2018-19), and Vice-President of the Student Council (2020).

Experience

End-of-Studies Internship

Toulouse, France

MIAT, INRAE

Apr - Sept 2023

- Modelled Anti-Poaching as a Partially Observable Stochastic Game.
- Implemented a simulator using PettingZoo, with various MARL algorithms using RLlib and PyTorch ([1],[2]).

M2 Project

Paris, France

MICS, CENTRALESUPÉLEC

Nov 2022 - Apr 2023

- Worked on the Roommate Matching problem under Rank-Envy-Free matching constraints [3].
- Established complexity results for such matchings, and studied their relation to existing notions.

M1 Internship

Paris, France

MICS, CENTRALESUPÉLEC

May - July 2022

- Worked on Epidemiological models to model the evolution of COVID Variants.
- Implemented numerical schemes and calibrated models using real-world data.

Visiting Student

Shantou, China

UNIVERSITY OF SHANTOU

May - July 2018

- Developed approaches based on genetic algorithms for modelling of Li-ion batteries in Electric Vehicles.
- Worked on data analysis and Experimental Design to minimise physical testing.

Teaching

Deep Learning

Toulouse, France

UNIVERSITY OF TOULOUSE

November 2024 - December 2025

- Created and taught a course on Deep Learning using PyTorch (Years 2024-25 and 2025-26).
- Course offered to final-year Masters' students in [Mathematical Modelling of Ecological Systems](#).
- Course projects: Auto-Encoders for Image Segmentation and training CNN Classifiers for real-world datasets.

Skills

Programming Languages

Python, C, C++17, MATLAB

Libraries

JAX, PyTorch, Ray/RLlib, BenchMARL, NumPy, Pandas, Matplotlib, Pytest, Sphinx

Other Software/Skills

LaTeX, Make, CMake, Bash, Git, CI/CD, Docker (Basics)

Spoken Languages

English (C1/C2), French (B2), Hindi, Telugu

Software

[Anti-Poaching Environment](#)

MIAT, INRAE

- Reference implementation of the Anti-Poaching Environment, implemented using PettingZoo ([1],[2]).
- Implemented MARL algorithms using RLLib to leverage multiple CPUs or GPU using PyTorch backend.
- Simulator implements a formally prescribed Partially Observable Stochastic Game (POSG) model.

[Multi-Adversarial Team Games](#)

MIAT, INRAE

- JAX-based, GPU-compatible implementation of the solver for Multi-Adversarial Team Games [4].
- Implementation scales to 4v9 games with 8 actions each, and can be vectorised to solve multiple games in parallel batches.
- Solver can be installed as a Python package.

[Deep Learning Course \(French\)](#)

UNIVERSITY OF TOULOUSE

- Slides and project material for Teaching course offered to Masters' students (CC-BY-4.0 License).
- Course introduces Perceptrons, Dense Networks, CNNs and RNNs, taught in PyTorch.

Publications

1. Maddila P, Eric C, Chabrier P, Sabbadin R, Vinyals M. APE: An Anti-poaching Multi-Agent Reinforcement Learning Benchmark. In Seventeenth European Workshop on Reinforcement Learning. 2024. Available: <https://openreview.net/forum?id=JSWRnHC93W>
2. Maddila P, Sabbadin R, Vinyals M. Learning in Competitive-Cooperative Games for Anti-Poaching. In Workshop on Learning in Games. 2024.
3. Coutance B, Maddila P, Wilczynski A. Rank-Envy-Freeness in roommate matchings. In 26th European Conference on Artificial Intelligence (ECAI 2023). 2023.
4. Maddila P, Sabbadin R, Vinyals M. Efficiently Computing Approximate Nash Equilibria in Multi-Adversarial Team Games. In The 25th International Conference on Autonomous Agents and Multi-Agent Systems. 2026. Available: <https://hal.science/view/index/docid/5450199>
5. Maddila P, Sabbadin R, Vinyals M. Approximating Nash Equilibria in Finite-Horizon Multi-Adversarial Team Markov Games. In Forty-Second Annual Conference on Uncertainty in Artificial Intelligence. 2026. Available: <https://openreview.net/forum?id=yXg85NFLkO>
6. Agrawal A, Maddila P, Zhang C, Perabathini B, Lasaulce S. Cooperative energy efficient resource allocation in fast fading interference networks. In 2019 International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob). 2019. pp. 276–281.
7. Bulusu SKC, Maddila P, Shaiek H, Roviras D. HPA linearization for FBMC-OQAM signals with fast convergence-digital predistortion. In 2019 16th International Symposium on Wireless Communication Systems (ISWCS). 2019. pp. 133–138.
8. Yun L, Maddila P, Gao L, Peng X, Niu X, Garg A, et al. An integrated framework for minimization of inter lithium-ion cell temperature differences and the total volume of the cell of battery pack for electric vehicles. Energy Storage. 2019;1: e41.
9. Yun L, Li W, Garg A, Maddila P, Gao L, Fan Z, et al. Maximization of extraction of Cadmium and Zinc during recycling of spent battery mix: An application of combined genetic programming and simulated annealing approach. Journal of Cleaner Production. 2019;218: 130–140.